Documentatie si raspunsuri TCP

Exercitiul 1:

Filtram dupa “http” si observam ca avem 2 pachete. Vedem ca prima transmite file-ul catre gaia.cs.umass.edu.

IP adress: 192.168.86.68

Source Port: 55639

153	0.147682	192.168.86.68	128.119.245.12	HTTP	1451	POST /wireshark-labs/lab3-1-reply.htm HTTP/1.1 (text/plain)
179	0.192625	128.119.245.12	192.168.86.68	HTTP	843	HTTP/1.1 200 OK (text/html)

```
    0100 .... = Version: 4
    .... 0101 = Header Length: 20 bytes (5)
  > Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
    Total Length: 1437
    Identification: 0x0000 (0)
  > Flags: 0x40, Don’t fragment
    ...0 0000 0000 0000 = Fragment Offset: 0
    Time to Live: 64
    Protocol: TCP (6)
    Header Checksum: 0xa8ea [validation disabled]
    [Header checksum status: Unverified]
    Source Address: 192.168.86.68
    Destination Address: 128.119.245.12
v Transmission Control Protocol, Src Port: 55639, Dst Port: 80, Seq: 152041, Ack: 1, Len: 1385
    Source Port: 55639
    Destination Port: 80
    [Stream index: 0]
    [Conversation completeness: Incomplete, DATA (15)]
    [TCP Segment Len: 1385]
    Sequence Number: 152041    (relative sequence number)
    Sequence Number (raw): 4236801228
    [Next Sequence Number: 153426    (relative sequence number)]
    Acknowledgment Number: 1    (relative ack number)
    Acknowledgment number (raw): 1068969753
    1000 .... = Header Length: 32 bytes (8)
<
0010  05 9d 00 00 40 00 40 06  a8 ea c0 a8 56 44 80 77   ····@·@· ····VD·w
0020  f5 0c d9 57 00 50 fc 88  74 cc 3f b7 2f 19 80 18   ···W·P·· t·?·/···
0030  08 0a bd 46 00 00 01 01  08 0a 2b 3f e4 e2 e9 48   ···F···· ··+?···H
0040  a2 69 20 75 70 20 77 69  74 68 20 74 68 65 20 64   ·i up wi th the d
0050  69 73 74 61 6e 74 20 73  6f 62 73 20 6f 66 20 74   istant s obs of t
0060  68 65 20 6d 69 73 65 72  61 62 6c 65 0d 0a 4d 6f   he miser able··Mo
0070  63 6b 20 54 75 72 74 6c  65 2e 0d 0a 0d 0a 20 20   ck Turtl e.····  
0080  53 6f 20 73 68 65 20 73  61 74 20 6f 6e 2c 20 77   So she s at on, w
0090  69 74 68 20 63 6c 6f 73  65 64 20 65 79 65 73 2c   ith clos ed eyes,
00a0  20 61 6e 64 20 68 61 6c  66 20 62 65 6c 69 65 76    and hal f believ
00b0  65 64 20 68 65 72 73 65  6c 66 20 69 6e 0d 0a 57   ed herse lf in··W
00c0  6f 6e 64 65 72 6c 61 6e  64 2c 20 74 68 6f 75 67   onderlan d, thoug
00d0  68 20 73 68 65 20 6b 6e  65 77 20 73 68 65 20 68   h she kn ew she h
00e0  61 64 20 62 75 74 20 74  6f 20 6f 70 65 6e 20 74   ad but t o open t
00f0  68 65 6d 20 61 67 61 69  6e 2c 20 61 6e 64 0d 0a   hem agai n, and··
0100  61 6c 6c 20 77 6f 75 6c  64 20 63 68 61 6e 67 65   all woul d change
0110  20 74 6f 20 64 75 6c 6c  20 72 65 61 6c 69 74 79    to dull  reality
0120  2d 2d 74 68 65 20 67 72  61 73 73 20 77 6f 75 6c   --the gr ass woul
```

Exercitiul 2:

Pastram filtrul folosit la exercitiul anterior si observam mai jos IP address ul si portul.

IP address of gaia.cs.umass.edu: 128.119.245.12

Destination Port: 80

153	0.147682	192.168.86.68	128.119.245.12	HTTP	1451	POST /wireshark-labs/lab3-1-reply.htm HTTP/1.1 (text/plain)
179	0.192625	128.119.245.12	192.168.86.68	HTTP	843	HTTP/1.1 200 OK (text/html)

```
    0100 .... = Version: 4
    .... 0101 = Header Length: 20 bytes (5)
  > Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
    Total Length: 1437
    Identification: 0x0000 (0)
  > Flags: 0x40, Don't fragment
    ...0 0000 0000 0000 = Fragment Offset: 0
    Time to Live: 64
    Protocol: TCP (6)
    Header Checksum: 0xa8ea [validation disabled]
    [Header checksum status: Unverified]
    Source Address: 192.168.86.68
    Destination Address: 128.119.245.12
v Transmission Control Protocol, Src Port: 55639, Dst Port: 80, Seq: 152041, Ack: 1, Len: 1385
    Source Port: 55639
    Destination Port: 80
    [Stream index: 0]
    [Conversation completeness: Incomplete, DATA (15)]
    [TCP Segment Len: 1385]
    Sequence Number: 152041    (relative sequence number)
    Sequence Number (raw): 4236801228
    [Next Sequence Number: 153426    (relative sequence number)]
    Acknowledgment Number: 1    (relative ack number)
    Acknowledgment number (raw): 1068969753
    1000 .... = Header Length: 32 bytes (8)
```

```
0010  05 9d 00 00 40 00 40 06  a8 ea c0 a8 56 44 80 77   ····@·@· ····VD·w
0020  f5 0c d9 57 00 50 fc 88  74 cc 3f b7 2f 19 80 18   ···W·P·· t·?·/···
0030  08 0a bd 46 00 00 01 01  08 0a 2b 3f e4 e2 e9 48   ···F···· ··+?···H
0040  a2 69 20 75 70 20 77 69  74 68 20 74 68 65 20 64   ·i up wi th the d
0050  69 73 74 61 6e 74 20 73  6f 62 73 20 6f 66 20 74   istant s obs of t
0060  68 65 20 6d 69 73 65 72  61 62 6c 65 0d 0a 4d 6f   he miser able··Mo
0070  63 6b 20 54 75 72 74 6c  65 2e 0d 0a 0d 0a 20 20   ck Turtl e.····
0080  53 6f 20 73 68 65 20 73  61 74 20 6f 6e 2c 20 77   So she s at on, w
0090  69 74 68 20 63 6c 6f 73  65 64 20 65 79 65 73 2c   ith clos ed eyes,
00a0  20 61 6e 64 20 68 61 6c  66 20 62 65 6c 69 65 76    and hal f believ
00b0  65 64 20 68 65 72 73 65  6c 66 20 69 6e 0d 0a 57   ed herse lf in··W
00c0  6f 6e 64 65 72 6c 61 6e  64 2c 20 74 68 6f 75 67   onderlan d, thoug
00d0  68 20 73 68 65 20 6b 6e  65 77 20 73 68 65 20 68   h she kn ew she h
00e0  61 64 20 62 75 74 20 74  6f 20 6f 70 65 6e 20 74   ad but t o open t
00f0  68 65 6d 20 61 67 61 69  6e 2c 20 61 6e 64 0d 0a   hem agai n, and··
0100  61 6c 6c 20 77 6f 75 6c  64 20 63 68 61 6e 67 65   all woul d change
0110  20 74 6f 20 64 75 6c 6c  20 72 65 61 6c 69 74 79    to dull  reality
0120  2d 2d 74 68 65 20 67 72  61 73 73 20 77 6f 75 6c   --the gr ass woul
```

Exercitiul 3:

Filtram dupa “tcp” si intram pe primul pachet.

Sequence number of TCP: 4236649187

Flagul de SYN este setat.

Selective Acknowledgements este permis.

1	0.000000	192.168.86.68	128.119.245.12	TCP	78	55639 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=64 TSval=725607509 TSecr=0 SACK_…
2	0.022414	128.119.245.12	192.168.86.68	TCP	74	80 → 55639 [SYN, ACK] Seq=0 Ack=1 Win=28960 Len=0 MSS=1460 SACK_PERM=1 TSval=391385…
3	0.022505	192.168.86.68	128.119.245.12	TCP	66	55639 → 80 [ACK] Seq=1 Ack=1 Win=131712 Len=0 TSval=725607531 TSecr=3913851370
4	0.024047	192.168.86.68	128.119.245.12	TCP	1514	55639 → 80 [ACK] Seq=1 Ack=1 Win=131712 Len=1448 TSval=725607532 TSecr=3913851370 […
5	0.024048	192.168.86.68	128.119.245.12	TCP	1514	55639 → 80 [ACK] Seq=1449 Ack=1 Win=131712 Len=1448 TSval=725607532 TSecr=391385137…
6	0.024049	192.168.86.68	128.119.245.12	TCP	1514	55639 → 80 [ACK] Seq=2897 Ack=1 Win=131712 Len=1448 TSval=725607532 TSecr=391385137…
7	0.052671	128.119.245.12	192.168.86.68	TCP	66	80 → 55639 [ACK] Seq=1 Ack=1449 Win=31872 Len=0 TSval=3913851399 TSecr=725607532
8	0.052676	128.119.245.12	192.168.86.68	TCP	66	80 → 55639 [ACK] Seq=1 Ack=2897 Win=34816 Len=0 TSval=3913851400 TSecr=725607532
9	0.052774	192.168.86.68	128.119.245.12	TCP	1514	55639 → 80 [ACK] Seq=4345 Ack=1 Win=131712 Len=1448 TSval=725607560 TSecr=391385139…
10	0.052775	192.168.86.68	128.119.245.12	TCP	1514	55639 → 80 [ACK] Seq=5793 Ack=1 Win=131712 Len=1448 TSval=725607560 TSecr=391385139…
11	0.052854	192.168.86.68	128.119.245.12	TCP	1514	55639 → 80 [ACK] Seq=7241 Ack=1 Win=131712 Len=1448 TSval=725607560 TSecr=391385140…
12	0.052855	192.168.86.68	128.119.245.12	TCP	1514	55639 → 80 [ACK] Seq=8689 Ack=1 Win=131712 Len=1448 TSval=725607560 TSecr=391385140…
13	0.053626	128.119.245.12	192.168.86.68	TCP	66	80 → 55639 [ACK] Seq=1 Ack=4345 Win=37760 Len=0 TSval=3913851400 TSecr=725607532
14	0.053710	192.168.86.68	128.119.245.12	TCP	1514	55639 → 80 [ACK] Seq=10137 Ack=1 Win=131712 Len=1448 TSval=725607560 TSecr=39138514…
15	0.053711	192.168.86.68	128.119.245.12	TCP	1514	55639 → 80 [ACK] Seq=11585 Ack=1 Win=131712 Len=1448 TSval=725607560 TSecr=39138514…
16	0.080768	128.119.245.12	192.168.86.68	TCP	66	80 → 55639 [ACK] Seq=1 Ack=5793 Win=40576 Len=0 TSval=3913851421 TSecr=725607560
17	0.080771	128.119.245.12	192.168.86.68	TCP	66	80 → 55639 [ACK] Seq=1 Ack=7241 Win=43520 Len=0 TSval=3913851422 TSecr=725607560
18	0.080772	128.119.245.12	192.168.86.68	TCP	66	80 → 55639 [ACK] Seq=1 Ack=8689 Win=46336 Len=0 TSval=3913851422 TSecr=725607560
19	0.080772	128.119.245.12	192.168.86.68	TCP	66	80 → 55639 [ACK] Seq=1 Ack=10137 Win=49280 Len=0 TSval=3913851422 TSecr=725607560
20	0.080845	192.168.86.68	128.119.245.12	TCP	1514	55639 → 80 [ACK] Seq=13033 Ack=1 Win=131712 Len=1448 TSval=725607588 TSecr=39138514…
21	0.080846	192.168.86.68	128.119.245.12	TCP	1514	55639 → 80 [ACK] Seq=14481 Ack=1 Win=131712 Len=1448 TSval=725607588 TSecr=39138514…
22	0.080847	192.168.86.68	128.119.245.12	TCP	1514	55639 → 80 [ACK] Seq=15929 Ack=1 Win=131712 Len=1448 TSval=725607588 TSecr=39138514…

```
Sequence Number: 0    (relative sequence number)
Sequence Number (raw): 4236649187
[Next Sequence Number: 1    (relative sequence number)]
Acknowledgment Number: 0
Acknowledgment number (raw): 0
1011 .... = Header Length: 44 bytes (11)
Flags: 0x002 (SYN)
    000. .... .... = Reserved: Not set
    ...0 .... .... = Nonce: Not set
    .... 0... .... = Congestion Window Reduced (CWR): Not set
    .... .0.. .... = ECN-Echo: Not set
    .... ..0. .... = Urgent: Not set
    .... ...0 .... = Acknowledgment: Not set
    .... .... 0... = Push: Not set
    .... .... .0.. = Reset: Not set
  > .... .... ..1. = Syn: Set
    .... .... ...0 = Fin: Not set
    [TCP Flags: ··********S·]
Window: 65535
[Calculated window size: 65535]
Checksum: 0xa1e4 [unverified]
[Checksum Status: Unverified]
Urgent Pointer: 0
Options: (24 bytes), Maximum segment size, No-Operation (NOP), Window scale, No-Operation (NOP), No-Operation (NOP), Timestamps, SACK permitted, End of Option List (EOL)
  > TCP Option - Maximum segment size: 1460 bytes
  > TCP Option - No-Operation (NOP)
  > TCP Option - Window scale: 6 (multiply by 64)
  > TCP Option - No-Operation (NOP)
  > TCP Option - No-Operation (NOP)
  > TCP Option - Timestamps: TSval 725607509, TSecr 0
  > TCP Option - SACK permitted
  > TCP Option - End of Option List (EOL)
> [Timestamps]
```

```
0000  3c 28 6d 89 0e c8 78 4f  43 98 d9 27 08 00 45 00   <(m···xO C··'··E·
0010  00 40 00 00 40 00 40 06  ae 47 c0 a8 56 44 80 77   ·@··@·@·  ·G··VD·w
0020  f5 0c d9 57 00 50 fc 86  22 e3 00 00 00 00 b0 02   ···W·P·· "·******
0030  ff ff a1 e4 00 00 02 04  05 b4 01 03 03 06 01 01   ********  ********
0040  08 0a 2b 3f e4 55 00 00  00 00 04 02 00 00         ··+?·U·· ******
```

Exercitiul 4:

Pastram filtrul de “tcp” si intram pe al doilea pachet.

Sequence number SYNACK: 1068969752

Segmentul este SYNACK deoarece are flaguri SYN si ACK setate.

Valoarea Acknowledgement numberul este: 4236649188. Aceasta valoare a fost determinata prin adaugarea valorii 1 la sequence numberul de la SYN.

1	0.000000	192.168.86.68	128.119.245.12	TCP	78	55639 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=64 TSval=725607509 TSecr=0 S
2	0.022414	128.119.245.12	192.168.86.68	TCP	74	80 → 55639 [SYN, ACK] Seq=0 Ack=1 Win=28960 Len=0 MSS=1460 SACK_PERM=1 TSval=39
3	0.022505	192.168.86.68	128.119.245.12	TCP	66	55639 → 80 [ACK] Seq=1 Ack=1 Win=131712 Len=0 TSval=725607531 TSecr=3913851370
4	0.024047	192.168.86.68	128.119.245.12	TCP	1514	55639 → 80 [ACK] Seq=1 Ack=1 Win=131712 Len=1448 TSval=725607532 TSecr=39138513
5	0.024048	192.168.86.68	128.119.245.12	TCP	1514	55639 → 80 [ACK] Seq=1449 Ack=1 Win=131712 Len=1448 TSval=725607532 TSecr=39138
6	0.024049	192.168.86.68	128.119.245.12	TCP	1514	55639 → 80 [ACK] Seq=2897 Ack=1 Win=131712 Len=1448 TSval=725607532 TSecr=39138
7	0.052671	128.119.245.12	192.168.86.68	TCP	66	80 → 55639 [ACK] Seq=1 Ack=1449 Win=31872 Len=0 TSval=3913851399 TSecr=72560753
8	0.052676	128.119.245.12	192.168.86.68	TCP	66	80 → 55639 [ACK] Seq=1 Ack=2897 Win=34816 Len=0 TSval=3913851400 TSecr=72560753
9	0.052774	192.168.86.68	128.119.245.12	TCP	1514	55639 → 80 [ACK] Seq=4345 Ack=1 Win=131712 Len=1448 TSval=725607560 TSecr=39138
10	0.052775	192.168.86.68	128.119.245.12	TCP	1514	55639 → 80 [ACK] Seq=5793 Ack=1 Win=131712 Len=1448 TSval=725607560 TSecr=39138
11	0.052854	192.168.86.68	128.119.245.12	TCP	1514	55639 → 80 [ACK] Seq=7241 Ack=1 Win=131712 Len=1448 TSval=725607560 TSecr=39138
12	0.052855	192.168.86.68	128.119.245.12	TCP	1514	55639 → 80 [ACK] Seq=8689 Ack=1 Win=131712 Len=1448 TSval=725607560 TSecr=39138
13	0.053626	128.119.245.12	192.168.86.68	TCP	66	80 → 55639 [ACK] Seq=1 Ack=4345 Win=37760 Len=0 TSval=3913851400 TSecr=72560753
14	0.053710	192.168.86.68	128.119.245.12	TCP	1514	55639 → 80 [ACK] Seq=10137 Ack=1 Win=131712 Len=1448 TSval=725607560 TSecr=3913
15	0.053711	192.168.86.68	128.119.245.12	TCP	1514	55639 → 80 [ACK] Seq=11585 Ack=1 Win=131712 Len=1448 TSval=725607560 TSecr=3913
16	0.080768	128.119.245.12	192.168.86.68	TCP	66	80 → 55639 [ACK] Seq=1 Ack=5793 Win=40576 Len=0 TSval=3913851421 TSecr=72560756
17	0.080771	128.119.245.12	192.168.86.68	TCP	66	80 → 55639 [ACK] Seq=1 Ack=7241 Win=43520 Len=0 TSval=3913851422 TSecr=72560756
18	0.080772	128.119.245.12	192.168.86.68	TCP	66	80 → 55639 [ACK] Seq=1 Ack=8689 Win=46336 Len=0 TSval=3913851422 TSecr=72560756
19	0.080772	128.119.245.12	192.168.86.68	TCP	66	80 → 55639 [ACK] Seq=1 Ack=10137 Win=49280 Len=0 TSval=3913851422 TSecr=7256075
20	0.080845	192.168.86.68	128.119.245.12	TCP	1514	55639 → 80 [ACK] Seq=13033 Ack=1 Win=131712 Len=1448 TSval=725607588 TSecr=3913
21	0.080846	192.168.86.68	128.119.245.12	TCP	1514	55639 → 80 [ACK] Seq=14481 Ack=1 Win=131712 Len=1448 TSval=725607588 TSecr=3913
22	0.080847	192.168.86.68	128.119.245.12	TCP	1514	55639 → 80 [ACK] Seq=15929 Ack=1 Win=131712 Len=1448 TSval=725607588 TSecr=3913

```
    Source Address: 128.119.245.12
    Destination Address: 192.168.86.68
v Transmission Control Protocol, Src Port: 80, Dst Port: 55639, Seq: 0, Ack: 1, Len: 0
    Source Port: 80
    Destination Port: 55639
    [Stream index: 0]
    [Conversation completeness: Incomplete, DATA (15)]
    [TCP Segment Len: 0]
    Sequence Number: 0    (relative sequence number)
    Sequence Number (raw): 1068969752
    [Next Sequence Number: 1    (relative sequence number)]
    Acknowledgment Number: 1    (relative ack number)
    Acknowledgment number (raw): 4236649188
    1010 .... = Header Length: 40 bytes (10)
  v Flags: 0x012 (SYN, ACK)
      000. .... .... = Reserved: Not set
      ...0 .... .... = Nonce: Not set
      .... 0... .... = Congestion Window Reduced (CWR): Not set
      .... .0.. .... = ECN-Echo: Not set
      .... ..0. .... = Urgent: Not set
      .... ...1 .... = Acknowledgment: Set
      .... .... 0... = Push: Not set
      .... .... .0.. = Reset: Not set
    > .... .... ..1. = Syn: Set
      .... .... ...0 = Fin: Not set
      [TCP Flags: ·******A··S·]
    Window: 28960
    [Calculated window size: 28960]
    Checksum: 0x47b4 [unverified]
    [Checksum Status: Unverified]
    Urgent Pointer: 0
  v Options: (20 bytes), Maximum segment size, SACK permitted, Timestamps, No-Operation (NOP), Window scale
    > TCP Option - Maximum segment size: 1460 bytes
    > TCP Option - SACK permitted
```

```
0000  78 4f 43 98 d9 27 3c 28  6d 89 0e c8 08 00 45 00   xOC··'<( m·····E·
0010  00 3c 00 00 40 00 34 06  ba 4b 80 77 f5 0c c0 a8   ·<··@·4· ·K·w····
0020  56 44 00 50 d9 57 3f b7  2f 18 fc 86 22 e4 a0 12   VD·P·W?· /···"···
0030  71 20 47 b4 00 00 02 04  05 b4 04 02 08 0a e9 48   q G····· *******H
0040  a1 ea 2b 3f e4 55 01 03  03 07                     ··+?·U·· ··
```

Exercitiul 5:

Pastram filtrul “tcp” si observam ca in a patrulea pachet avem in ASCII POST (cum se spunea in cerinta).

Sequence number: 4236649188.

Payload (data) contine 1448 bytes.

Nu toate datele au fost transferate, putem observa in urmatoarele pachete ca inca se trimite file-ul.

tcp

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	192.168.86.68	128.119.245.12	TCP	78	55639 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=64 TSval=
2	0.022414	128.119.245.12	192.168.86.68	TCP	74	80 → 55639 [SYN, ACK] Seq=0 Ack=1 Win=28960 Len=0 MSS=1460 S
3	0.022505	192.168.86.68	128.119.245.12	TCP	66	55639 → 80 [ACK] Seq=1 Ack=1 Win=131712 Len=0 TSval=7256075
4	0.024047	192.168.86.68	128.119.245.12	TCP	1514	55639 → 80 [ACK] Seq=1 Ack=1 Win=131712 Len=1448 TSval=72560
5	0.024048	192.168.86.68	128.119.245.12	TCP	1514	55639 → 80 [ACK] Seq=1449 Ack=1 Win=131712 Len=1448 TSval=72
6	0.024049	192.168.86.68	128.119.245.12	TCP	1514	55639 → 80 [ACK] Seq=2897 Ack=1 Win=131712 Len=1448 TSval=72
7	0.052671	128.119.245.12	192.168.86.68	TCP	66	80 → 55639 [ACK] Seq=1 Ack=1449 Win=31872 Len=0 TSval=391385
8	0.052676	128.119.245.12	192.168.86.68	TCP	66	80 → 55639 [ACK] Seq=1 Ack=2897 Win=34816 Len=0 TSval=391385
9	0.052774	192.168.86.68	128.119.245.12	TCP	1514	55639 → 80 [ACK] Seq=4345 Ack=1 Win=131712 Len=1448 TSval=72
10	0.052775	192.168.86.68	128.119.245.12	TCP	1514	55639 → 80 [ACK] Seq=5793 Ack=1 Win=131712 Len=1448 TSval=72
11	0.052854	192.168.86.68	128.119.245.12	TCP	1514	55639 → 80 [ACK] Seq=7241 Ack=1 Win=131712 Len=1448 TSval=72
12	0.052855	192.168.86.68	128.119.245.12	TCP	1514	55639 → 80 [ACK] Seq=8689 Ack=1 Win=131712 Len=1448 TSval=72
13	0.053626	128.119.245.12	192.168.86.68	TCP	66	80 → 55639 [ACK] Seq=1 Ack=4345 Win=37760 Len=0 TSval=391385
14	0.053710	192.168.86.68	128.119.245.12	TCP	1514	55639 → 80 [ACK] Seq=10137 Ack=1 Win=131712 Len=1448 TSval=7
15	0.053711	192.168.86.68	128.119.245.12	TCP	1514	55639 → 80 [ACK] Seq=11585 Ack=1 Win=131712 Len=1448 TSval=7
16	0.080768	128.119.245.12	192.168.86.68	TCP	66	80 → 55639 [ACK] Seq=1 Ack=5793 Win=40576 Len=0 TSval=391385
17	0.080771	128.119.245.12	192.168.86.68	TCP	66	80 → 55639 [ACK] Seq=1 Ack=7241 Win=43520 Len=0 TSval=391385
18	0.080772	128.119.245.12	192.168.86.68	TCP	66	80 → 55639 [ACK] Seq=1 Ack=8689 Win=46336 Len=0 TSval=391385
19	0.080772	128.119.245.12	192.168.86.68	TCP	66	80 → 55639 [ACK] Seq=1 Ack=10137 Win=49280 Len=0 TSval=39138
20	0.080845	192.168.86.68	128.119.245.12	TCP	1514	55639 → 80 [ACK] Seq=13033 Ack=1 Win=131712 Len=1448 TSval=7
21	0.080846	192.168.86.68	128.119.245.12	TCP	1514	55639 → 80 [ACK] Seq=14481 Ack=1 Win=131712 Len=1448 TSval=7
22	0.080847	192.168.86.68	128.119.245.12	TCP	1514	55639 → 80 [ACK] Seq=15929 Ack=1 Win=131712 Len=1448 TSval=7

```
    [TCP Segment Len: 1448]
    Sequence Number: 1    (relative sequence number)
    Sequence Number (raw): 4236649188
    [Next Sequence Number: 1449    (relative sequence number)]
    Acknowledgment Number: 1    (relative ack number)
    Acknowledgment number (raw): 1068969753
    1000 .... = Header Length: 32 bytes (8)
  ⌄ Flags: 0x010 (ACK)
      000. .... .... = Reserved: Not set
      ...0 .... .... = Nonce: Not set
      .... 0... .... = Congestion Window Reduced (CWR): Not set
      .... .0.. .... = ECN-Echo: Not set
      .... ..0. .... = Urgent: Not set
      .... ...1 .... = Acknowledgment: Set
      .... .... 0... = Push: Not set
      .... .... .0.. = Reset: Not set
      .... .... ..0. = Syn: Not set
      .... .... ...0 = Fin: Not set
      [TCP Flags: ·======·A····]
    Window: 2058
    [Calculated window size: 131712]
    [Window size scaling factor: 64]
    Checksum: 0xbd21 [unverified]
    [Checksum Status: Unverified]
    Urgent Pointer: 0
  ⌄ Options: (12 bytes), No-Operation (NOP), No-Operation (NOP), Timestamps
    > TCP Option - No-Operation (NOP)
    > TCP Option - No-Operation (NOP)
    > TCP Option - Timestamps: TSval 725607532, TSecr 3913851370
  > [Timestamps]
  > [SEQ/ACK analysis]
    TCP payload (1448 bytes)
    [Reassembled PDU in frame: 153]
    TCP segment data (1448 bytes)
```

```
0020  f5 0c d9 57 00 50 fc 86  22 e4 3f b7 2f 19 80 10   ···W·P·· "·?·/···
0030  08 0a bd 21 00 00 01 01  08 0a 2b 3f e4 6c e9 48   ···!···· ··+?·l·H
0040  a1 ea 50 4f 53 54 20 2f  77 69 72 65 73 68 61 72   ··POST / wireshar
0050  6b 2d 6c 61 62 73 2f 6c  61 62 33 2d 31 2d 72 65   k-labs/l ab3-1-re
0060  70 6c 79 2e 68 74 6d 20  48 54 54 50 2f 31 2e 31   ply.htm  HTTP/1.1
0070  0d 0a 48 6f 73 74 3a 20  67 61 69 61 2e 63 73 2e   ··Host:  gaia.cs.
0080  75 6d 61 73 73 2e 65 64  75 0d 0a 55 73 65 72 2d   umass.ed u··User-
0090  41 67 65 6e 74 3a 20 4d  6f 7a 69 6c 6c 61 2f 35   Agent: M ozilla/5
00a0  2e 30 20 28 4d 61 63 69  6e 74 6f 73 68 3b 20 49   .0 (Maci ntosh; I
00b0  6e 74 65 6c 20 4d 61 63  20 4f 53 20 58 20 31 30   ntel Mac  OS X 10
00c0  2e 31 35 3b 20 72 76 3a  38 35 2e 30 29 20 47 65   .15; rv: 85.0) Ge
00d0  63 6b 6f 2f 32 30 31 30  30 31 30 31 20 46 69 72   cko/2010 0101 Fir
00e0  65 66 6f 78 2f 38 35 2e  30 0d 0a 41 63 63 65 70   efox/85. 0··Accep
00f0  74 3a 20 74 65 78 74 2f  68 74 6d 6c 2c 61 70 70   t: text/ html,app
0100  6c 69 63 61 74 69 6f 6e  2f 78 68 74 6d 6c 2b 78   lication /xhtml+x
0110  6d 6c 2c 61 70 70 6c 69  63 61 74 69 6f 6e 2f 78   ml,appli cation/x
0120  6d 6c 3b 71 3d 30 2e 39  2c 69 6d 61 67 65 2f 77   ml;q=0.9 ,image/w
0130  65 62 70 2c 2a 2f 2a 3b  71 3d 30 2e 38 0d 0a 41   ebp,*/*; q=0.8··A
0140  63 63 65 70 74 2d 4c 61  6e 67 75 61 67 65 3a 20   ccept-La nguage:
```

Exercitiul 6:

Pastram filtrul de “tcp” si ne uitam la al patrulea pachet.

Primul segment este in pachetul 4 la timpul: 0.024047.

Primul ACK a fost primit in pachetul 7 la timpul: 0.052671.

Primul RTT este: 0.052671 - 0.024047 = 0.028624.

Al doilea RTT este: 0.052676 – 0.024048 = 0.028628.

EstimatedRTT este: EstimatedRTT <- (1 – alpha) EstimatedRTT + alpha R’

EstimatedRTT = 0.875 * 0.028624 + 0.125 * 0.028628 = 0.0286245.

1	0.000000	192.168.86.68	128.119.245.12	TCP	78	55639 → 80 [SYN] Seq=0 Win=655
2	0.022414	128.119.245.12	192.168.86.68	TCP	74	80 → 55639 [SYN, ACK] Seq=0 Ac
3	0.022505	192.168.86.68	128.119.245.12	TCP	66	55639 → 80 [ACK] Seq=1 Ack=1 W
4	0.024047	192.168.86.68	128.119.245.12	TCP	1514	55639 → 80 [ACK] Seq=1 Ack=1 W
5	0.024048	192.168.86.68	128.119.245.12	TCP	1514	55639 → 80 [ACK] Seq=1449 Ack=
6	0.024049	192.168.86.68	128.119.245.12	TCP	1514	55639 → 80 [ACK] Seq=2897 Ack=
7	0.052671	128.119.245.12	192.168.86.68	TCP	66	80 → 55639 [ACK] Seq=1 Ack=144
8	0.052676	128.119.245.12	192.168.86.68	TCP	66	80 → 55639 [ACK] Seq=1 Ack=289
9	0.052774	192.168.86.68	128.119.245.12	TCP	1514	55639 → 80 [ACK] Seq=4345 Ack=
10	0.052775	192.168.86.68	128.119.245.12	TCP	1514	55639 → 80 [ACK] Seq=5793 Ack=
11	0.052854	192.168.86.68	128.119.245.12	TCP	1514	55639 → 80 [ACK] Seq=7241 Ack=
12	0.052855	192.168.86.68	128.119.245.12	TCP	1514	55639 → 80 [ACK] Seq=8689 Ack=

Exercitiul 7:

Toate primele 4 pachete (pachetele cu nr 4, 5, 6, 9) au acelasi nr de Byes la header si la payload.

Primul data-carrying: 1448 + 32 = 1480 bytes

Al doinea data-carrying: 1448 + 32 = 1480 bytes

Al treilea data-carrying: 1448 + 32 = 1480 bytes

Al patrulea data-carrying: 1448 + 32 = 1480 bytes

Total: 1480 * 4 = 5920 bytes

tcp

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	192.168.86.68	128.119.245.12	TCP	78	55639 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=64 TSval=725607
2	0.022414	128.119.245.12	192.168.86.68	TCP	74	80 → 55639 [SYN, ACK] Seq=0 Ack=1 Win=28960 Len=0 MSS=1460 SACK_PE
3	0.022505	192.168.86.68	128.119.245.12	TCP	66	55639 → 80 [ACK] Seq=1 Ack=1 Win=131712 Len=0 TSval=725607531 TSec
4	0.024047	192.168.86.68	128.119.245.12	TCP	1514	55639 → 80 [ACK] Seq=1 Ack=1 Win=131712 Len=1448 TSval=725607532 T
5	0.024048	192.168.86.68	128.119.245.12	TCP	1514	55639 → 80 [ACK] Seq=1449 Ack=1 Win=131712 Len=1448 TSval=72560753
6	0.024049	192.168.86.68	128.119.245.12	TCP	1514	55639 → 80 [ACK] Seq=2897 Ack=1 Win=131712 Len=1448 TSval=72560753
7	0.052671	128.119.245.12	192.168.86.68	TCP	66	80 → 55639 [ACK] Seq=1 Ack=1449 Win=31872 Len=0 TSval=3913851399 T
8	0.052676	128.119.245.12	192.168.86.68	TCP	66	80 → 55639 [ACK] Seq=1 Ack=2897 Win=34816 Len=0 TSval=3913851400 T
9	0.052774	192.168.86.68	128.119.245.12	TCP	1514	55639 → 80 [ACK] Seq=4345 Ack=1 Win=131712 Len=1448 TSval=72560756
10	0.052775	192.168.86.68	128.119.245.12	TCP	1514	55639 → 80 [ACK] Seq=5793 Ack=1 Win=131712 Len=1448 TSval=72560756
11	0.052854	192.168.86.68	128.119.245.12	TCP	1514	55639 → 80 [ACK] Seq=7241 Ack=1 Win=131712 Len=1448 TSval=72560756
12	0.052855	192.168.86.68	128.119.245.12	TCP	1514	55639 → 80 [ACK] Seq=8689 Ack=1 Win=131712 Len=1448 TSval=72560756
13	0.053626	128.119.245.12	192.168.86.68	TCP	66	80 → 55639 [ACK] Seq=1 Ack=4345 Win=37760 Len=0 TSval=3913851400 T
14	0.053710	192.168.86.68	128.119.245.12	TCP	1514	55639 → 80 [ACK] Seq=10137 Ack=1 Win=131712 Len=1448 TSval=7256075
15	0.053711	192.168.86.68	128.119.245.12	TCP	1514	55639 → 80 [ACK] Seq=11585 Ack=1 Win=131712 Len=1448 TSval=7256075
16	0.080768	128.119.245.12	192.168.86.68	TCP	66	80 → 55639 [ACK] Seq=1 Ack=5793 Win=40576 Len=0 TSval=3913851421 T
17	0.080771	128.119.245.12	192.168.86.68	TCP	66	80 → 55639 [ACK] Seq=1 Ack=7241 Win=43520 Len=0 TSval=3913851422 T
18	0.080772	128.119.245.12	192.168.86.68	TCP	66	80 → 55639 [ACK] Seq=1 Ack=8689 Win=46336 Len=0 TSval=3913851422 T
19	0.080772	128.119.245.12	192.168.86.68	TCP	66	80 → 55639 [ACK] Seq=1 Ack=10137 Win=49280 Len=0 TSval=3913851422
20	0.080845	192.168.86.68	128.119.245.12	TCP	1514	55639 → 80 [ACK] Seq=13033 Ack=1 Win=131712 Len=1448 TSval=7256075
21	0.080846	192.168.86.68	128.119.245.12	TCP	1514	55639 → 80 [ACK] Seq=14481 Ack=1 Win=131712 Len=1448 TSval=7256075
22	0.080847	192.168.86.68	128.119.245.12	TCP	1514	55639 → 80 [ACK] Seq=15929 Ack=1 Win=131712 Len=1448 TSval=7256075

```
    Sequence Number: 1    (relative sequence number)
    Sequence Number (raw): 4236649188
    [Next Sequence Number: 1449    (relative sequence number)]
    Acknowledgment Number: 1    (relative ack number)
    Acknowledgment number (raw): 1068969753
    1000 .... = Header Length: 32 bytes (8)
  ⌄ Flags: 0x010 (ACK)
      000. .... .... = Reserved: Not set
      ...0 .... .... = Nonce: Not set
      .... 0... .... = Congestion Window Reduced (CWR): Not set
      .... .0.. .... = ECN-Echo: Not set
      .... ..0. .... = Urgent: Not set
      .... ...1 .... = Acknowledgment: Set
      .... .... 0... = Push: Not set
      .... .... .0.. = Reset: Not set
      .... .... ..0. = Syn: Not set
      .... .... ...0 = Fin: Not set
      [TCP Flags: ·····A····]
    Window: 2058
    [Calculated window size: 131712]
    [Window size scaling factor: 64]
    Checksum: 0xbd21 [unverified]
    [Checksum Status: Unverified]
    Urgent Pointer: 0
  ⌄ Options: (12 bytes), No-Operation (NOP), No-Operation (NOP), Timestamps
    > TCP Option - No-Operation (NOP)
    > TCP Option - No-Operation (NOP)
    > TCP Option - Timestamps: TSval 725607532, TSecr 3913851370
  > [Timestamps]
  > [SEQ/ACK analysis]
    TCP payload (1448 bytes)
    [Reassembled PDU in frame: 153]
    TCP segment data (1448 bytes)
```

Exercitiul 8:

Pastram filtrul “tcp” si ne uitem prin primele 4 raspunsuri de data-carying pachete dupa calculated window size (adica la pachetele 7, 8, 13, 16).

Primul pachet: 31872.

Al doilea pachet: 34816.

Al treilea pachet: 37760.

Al patrulea pachet: 40576.

Minimul este la pachetul 7: 31872.

tcp

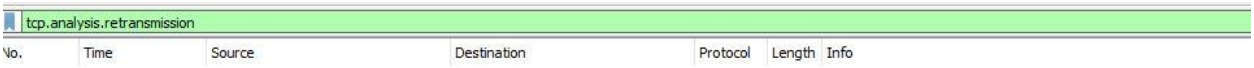
No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	192.168.86.68	128.119.245.12	TCP	78	55639 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=64 TSval=7256075
2	0.022414	128.119.245.12	192.168.86.68	TCP	74	80 → 55639 [SYN, ACK] Seq=0 Ack=1 Win=28960 Len=0 MSS=1460 SACK_PER
3	0.022505	192.168.86.68	128.119.245.12	TCP	66	55639 → 80 [ACK] Seq=1 Ack=1 Win=131712 Len=0 TSval=725607531 TSecr
4	0.024047	192.168.86.68	128.119.245.12	TCP	1514	55639 → 80 [ACK] Seq=1 Ack=1 Win=131712 Len=1448 TSval=725607532 TS
5	0.024048	192.168.86.68	128.119.245.12	TCP	1514	55639 → 80 [ACK] Seq=1449 Ack=1 Win=131712 Len=1448 TSval=725607532
6	0.024049	192.168.86.68	128.119.245.12	TCP	1514	55639 → 80 [ACK] Seq=2897 Ack=1 Win=131712 Len=1448 TSval=725607532
7	0.052671	128.119.245.12	192.168.86.68	TCP	66	80 → 55639 [ACK] Seq=1 Ack=1449 Win=31872 Len=0 TSval=3913851399 TS
8	0.052676	128.119.245.12	192.168.86.68	TCP	66	80 → 55639 [ACK] Seq=1 Ack=2897 Win=34816 Len=0 TSval=3913851400 TS
9	0.052774	192.168.86.68	128.119.245.12	TCP	1514	55639 → 80 [ACK] Seq=4345 Ack=1 Win=131712 Len=1448 TSval=725607560
10	0.052775	192.168.86.68	128.119.245.12	TCP	1514	55639 → 80 [ACK] Seq=5793 Ack=1 Win=131712 Len=1448 TSval=725607560
11	0.052854	192.168.86.68	128.119.245.12	TCP	1514	55639 → 80 [ACK] Seq=7241 Ack=1 Win=131712 Len=1448 TSval=725607560
12	0.052855	192.168.86.68	128.119.245.12	TCP	1514	55639 → 80 [ACK] Seq=8689 Ack=1 Win=131712 Len=1448 TSval=725607560
13	0.053626	128.119.245.12	192.168.86.68	TCP	66	80 → 55639 [ACK] Seq=1 Ack=4345 Win=37760 Len=0 TSval=3913851400 TS
14	0.053710	192.168.86.68	128.119.245.12	TCP	1514	55639 → 80 [ACK] Seq=10137 Ack=1 Win=131712 Len=1448 TSval=7256075
15	0.053711	192.168.86.68	128.119.245.12	TCP	1514	55639 → 80 [ACK] Seq=11585 Ack=1 Win=131712 Len=1448 TSval=7256075
16	0.080768	128.119.245.12	192.168.86.68	TCP	66	80 → 55639 [ACK] Seq=1 Ack=5793 Win=40576 Len=0 TSval=3913851421 TS
17	0.080771	128.119.245.12	192.168.86.68	TCP	66	80 → 55639 [ACK] Seq=1 Ack=7241 Win=43520 Len=0 TSval=3913851422 TS
18	0.080772	128.119.245.12	192.168.86.68	TCP	66	80 → 55639 [ACK] Seq=1 Ack=8689 Win=46336 Len=0 TSval=3913851422 TS
19	0.080772	128.119.245.12	192.168.86.68	TCP	66	80 → 55639 [ACK] Seq=1 Ack=10137 Win=49280 Len=0 TSval=3913851422 T
20	0.080845	192.168.86.68	128.119.245.12	TCP	1514	55639 → 80 [ACK] Seq=13033 Ack=1 Win=131712 Len=1448 TSval=7256075
21	0.080846	192.168.86.68	128.119.245.12	TCP	1514	55639 → 80 [ACK] Seq=14481 Ack=1 Win=131712 Len=1448 TSval=7256075
22	0.080847	192.168.86.68	128.119.245.12	TCP	1514	55639 → 80 [ACK] Seq=15929 Ack=1 Win=131712 Len=1448 TSval=7256075

```
    [Stream index: 0]
    [Conversation completeness: Incomplete, DATA (15)]
    [TCP Segment Len: 0]
    Sequence Number: 1    (relative sequence number)
    Sequence Number (raw): 1068969753
    [Next Sequence Number: 1    (relative sequence number)]
    Acknowledgment Number: 1449    (relative ack number)
    Acknowledgment number (raw): 4236650636
    1000 .... = Header Length: 32 bytes (8)
  ⌄ Flags: 0x010 (ACK)
      000. .... .... = Reserved: Not set
      ...0 .... .... = Nonce: Not set
      .... 0... .... = Congestion Window Reduced (CWR): Not set
      .... .0.. .... = ECN-Echo: Not set
      .... ..0. .... = Urgent: Not set
      .... ...1 .... = Acknowledgment: Set
      .... .... 0... = Push: Not set
      .... .... .0.. = Reset: Not set
      .... .... ..0. = Syn: Not set
      .... .... ...0 = Fin: Not set
      [TCP Flags: ·····A····]
    Window: 249
    [Calculated window size: 31872]
    [Window size scaling factor: 128]
    Checksum: 0xe0cb [unverified]
    [Checksum Status: Unverified]
    Urgent Pointer: 0
  ⌄ Options: (12 bytes), No-Operation (NOP), No-Operation (NOP), Timestamps
    > TCP Option - No-Operation (NOP)
    > TCP Option - No-Operation (NOP)
    > TCP Option - Timestamps: TSval 3913851399, TSecr 725607532
  > [Timestamps]
  > [SEQ/ACK analysis]
```

```
0000  78 4f 43 98 d9 27 3c 28  6d 89 0e c8 08 00 45 00   xOC··'<( m·····E·
0010  00 34 8d 1e 40 00 34 06  2d 35 80 77 f5 0c c0 a8   ·4··@·4· -5·w····
0020  56 44 00 50 d9 57 3f b7  2f 19 fc 86 28 8c 80 10   VD·P·W?· /···(···
0030  00 f9 e0 cb 00 00 01 01  08 0a e9 48 a2 07 2b 3f   ····· ···H··+?
0040  e4 6c                                              ·l
```

Exercitiul 9:

Filtram dupa “tcp.analysis.retransmission” si observam ca nu exista niciun pachet, deci nu au fost retransmise niciun segment.



Exercitiul 10:

Filtram dupa “tcp”.

Receiverul in mod tipic recunoaste 1448 de bytes, adica exact cat au fost trimis spre el, putem observa asta dupa ACKurile care se maresc cu cate 1448 intre ele in primele 10 pachete.

2	0.022414	128.119.245.12	192.168.86.68	TCP	74	80 → 55639 [SYN, ACK]
3	0.022505	192.168.86.68	128.119.245.12	TCP	66	55639 → 80 [ACK] Seq=
4	0.024047	192.168.86.68	128.119.245.12	TCP	1514	55639 → 80 [ACK] Seq=
5	0.024048	192.168.86.68	128.119.245.12	TCP	1514	55639 → 80 [ACK] Seq=
6	0.024049	192.168.86.68	128.119.245.12	TCP	1514	55639 → 80 [ACK] Seq=
7	0.052671	128.119.245.12	192.168.86.68	TCP	66	80 → 55639 [ACK] Seq=
8	0.052676	128.119.245.12	192.168.86.68	TCP	66	80 → 55639 [ACK] Seq=
9	0.052774	192.168.86.68	128.119.245.12	TCP	1514	55639 → 80 [ACK] Seq=
10	0.052775	192.168.86.68	128.119.245.12	TCP	1514	55639 → 80 [ACK] Seq=
11	0.052854	192.168.86.68	128.119.245.12	TCP	1514	55639 → 80 [ACK] Seq=
12	0.052855	192.168.86.68	128.119.245.12	TCP	1514	55639 → 80 [ACK] Seq=
13	0.053626	128.119.245.12	192.168.86.68	TCP	66	80 → 55639 [ACK] Seq=
14	0.053710	192.168.86.68	128.119.245.12	TCP	1514	55639 → 80 [ACK] Seq=
15	0.053711	192.168.86.68	128.119.245.12	TCP	1514	55639 → 80 [ACK] Seq=
16	0.080768	128.119.245.12	192.168.86.68	TCP	66	80 → 55639 [ACK] Seq=
17	0.080771	128.119.245.12	192.168.86.68	TCP	66	80 → 55639 [ACK] Seq=
18	0.080772	128.119.245.12	192.168.86.68	TCP	66	80 → 55639 [ACK] Seq=

```
    Destination Address: 192.168.86.68
  ✓ Transmission Control Protocol, Src Port: 80, Dst Port: 55639, Seq: 0, Ack: 1, Len: 0
    Source Port: 80
```

7	0.052671	128.119.245.12	192.168.86.68	TCP	66	80 → 55639 [ACK] S
8	0.052676	128.119.245.12	192.168.86.68	TCP	66	80 → 55639 [ACK] S
9	0.052774	192.168.86.68	128.119.245.12	TCP	1514	55639 → 80 [ACK] S
10	0.052775	192.168.86.68	128.119.245.12	TCP	1514	55639 → 80 [ACK] S
11	0.052854	192.168.86.68	128.119.245.12	TCP	1514	55639 → 80 [ACK] S
12	0.052855	192.168.86.68	128.119.245.12	TCP	1514	55639 → 80 [ACK] S
13	0.053626	128.119.245.12	192.168.86.68	TCP	66	80 → 55639 [ACK] S
14	0.053710	192.168.86.68	128.119.245.12	TCP	1514	55639 → 80 [ACK] S
15	0.053711	192.168.86.68	128.119.245.12	TCP	1514	55639 → 80 [ACK] S
16	0.080768	128.119.245.12	192.168.86.68	TCP	66	80 → 55639 [ACK] S
17	0.080771	128.119.245.12	192.168.86.68	TCP	66	80 → 55639 [ACK] S
18	0.080772	128.119.245.12	192.168.86.68	TCP	66	80 → 55639 [ACK] S

```
    Destination Address: 192.168.86.68
  ✓ Transmission Control Protocol, Src Port: 80, Dst Port: 55639, Seq: 1, Ack: 1449, Len: 0
    Source Port: 80
    Destination Port: 55639
```

8	0.052676	128.119.245.12	192.168.86.68	TCP	66	80 → 55639 [ACK] Seq=1 Ack=2
9	0.052774	192.168.86.68	128.119.245.12	TCP	1514	55639 → 80 [ACK] Seq=4345 Ac
10	0.052775	192.168.86.68	128.119.245.12	TCP	1514	55639 → 80 [ACK] Seq=5793 Ac
11	0.052854	192.168.86.68	128.119.245.12	TCP	1514	55639 → 80 [ACK] Seq=7241 Ac
12	0.052855	192.168.86.68	128.119.245.12	TCP	1514	55639 → 80 [ACK] Seq=8689 Ac
13	0.053626	128.119.245.12	192.168.86.68	TCP	66	80 → 55639 [ACK] Seq=1 Ack=4
14	0.053710	192.168.86.68	128.119.245.12	TCP	1514	55639 → 80 [ACK] Seq=10137 A
15	0.053711	192.168.86.68	128.119.245.12	TCP	1514	55639 → 80 [ACK] Seq=11585 A
16	0.080768	128.119.245.12	192.168.86.68	TCP	66	80 → 55639 [ACK] Seq=1 Ack=5
17	0.080771	128.119.245.12	192.168.86.68	TCP	66	80 → 55639 [ACK] Seq=1 Ack=7
18	0.080772	128.119.245.12	192.168.86.68	TCP	66	80 → 55639 [ACK] Seq=1 Ack=8

```
    Destination Address: 192.168.86.68
  ✓ Transmission Control Protocol, Src Port: 80, Dst Port: 55639, Seq: 1, Ack: 2897, Len: 0
    Source Port: 80
```

Exercitiul 11:

Throughput (bytes transferred per unit time) for the TCP connection se calculeaza prin volumul de date trimis in bytes pe RTT aferent.

Volumul de date trimis este marimea cartii care este: 148 kB.

Primul send este la pachetul 4 a timpul: 0.024047.

Ultimul receive a fost la pachetul cu numarul 178 la timpul: 0.191496.

RTT toata cartea trimisa: 0.191496 - 0.024047 = 0.167449.

Throughput (bytes transferred per unit time) for the TCP connection este:

148 kB / 0.167449 sec = 883.851202456 kB/sec.

tcp

No.	Time	Source	Destination
1	0.000000	192.168.86.68	128.119.245.12
2	0.022414	128.119.245.12	192.168.86.68
3	0.022505	192.168.86.68	128.119.245.12
4	0.024047	192.168.86.68	128.119.245.12
5	0.024048	192.168.86.68	128.119.245.12

primul send

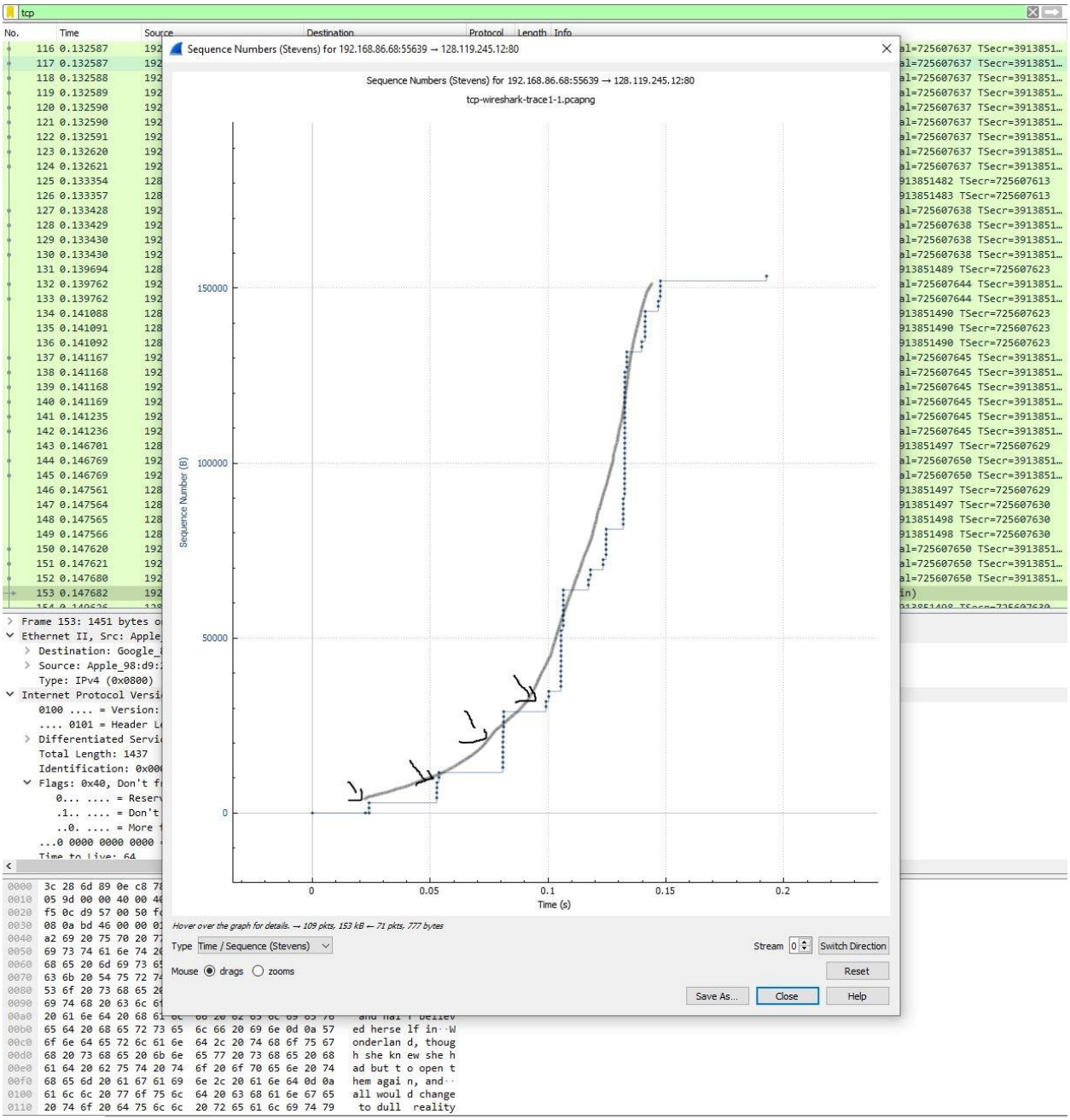
tcp

No.	Time	Source	Destination	Protocol	Length	Info
164	0.157197	128.119.245.12	192.168.86.68	TCP	66	80 → 55639 [ACK] Seq=1 Ack=110049 Win=183296 Len=0 TSval=3913851507 TSe
165	0.167827	128.119.245.12	192.168.86.68	TCP	66	80 → 55639 [ACK] Seq=1 Ack=112945 Win=183296 Len=0 TSval=3913851508 TSe
166	0.167832	128.119.245.12	192.168.86.68	TCP	66	80 → 55639 [ACK] Seq=1 Ack=117289 Win=183296 Len=0 TSval=3913851513 TSe
167	0.167833	128.119.245.12	192.168.86.68	TCP	66	80 → 55639 [ACK] Seq=1 Ack=123081 Win=183296 Len=0 TSval=3913851513 TSe
168	0.167835	128.119.245.12	192.168.86.68	TCP	66	80 → 55639 [ACK] Seq=1 Ack=127425 Win=183296 Len=0 TSval=3913851513 TSe
169	0.167836	128.119.245.12	192.168.86.68	TCP	66	80 → 55639 [ACK] Seq=1 Ack=130321 Win=183296 Len=0 TSval=3913851514 TSe
170	0.167837	128.119.245.12	192.168.86.68	TCP	66	80 → 55639 [ACK] Seq=1 Ack=134665 Win=183296 Len=0 TSval=3913851514 TSe
171	0.167839	128.119.245.12	192.168.86.68	TCP	66	80 → 55639 [ACK] Seq=1 Ack=137561 Win=183296 Len=0 TSval=3913851514 TSe
172	0.167840	128.119.245.12	192.168.86.68	TCP	66	80 → 55639 [ACK] Seq=1 Ack=140457 Win=183296 Len=0 TSval=3913851516 TSe
173	0.167841	128.119.245.12	192.168.86.68	TCP	66	80 → 55639 [ACK] Seq=1 Ack=143353 Win=183296 Len=0 TSval=3913851516 TSe
174	0.174624	128.119.245.12	192.168.86.68	TCP	66	80 → 55639 [ACK] Seq=1 Ack=146249 Win=183296 Len=0 TSval=3913851524 TSe
175	0.175804	128.119.245.12	192.168.86.68	TCP	66	80 → 55639 [ACK] Seq=1 Ack=149145 Win=186112 Len=0 TSval=3913851524 TSe
176	0.185113	128.119.245.12	192.168.86.68	TCP	66	80 → 55639 [ACK] Seq=1 Ack=150593 Win=189056 Len=0 TSval=3913851535 TSe
177	0.191491	128.119.245.12	192.168.86.68	TCP	66	80 → 55639 [ACK] Seq=1 Ack=152041 Win=191872 Len=0 TSval=3913851541 TSe
178	0.191496	128.119.245.12	192.168.86.68	TCP	66	80 → 55639 [ACK] Seq=1 Ack=153426 Win=194816 Len=0 TSval=3913851541 TSe
179	0.192625	128.119.245.12	192.168.86.68	HTTP	843	HTTP/1.1 200 OK (text/html)
180	0.192732	192.168.86.68	128.119.245.12	TCP	66	55639 → 80 [ACK] Seq=153426 Ack=778 Win=130944 Len=0 TSval=725607691 TS

last receive

Exercitiul 12:

Plotam graficul cu ajutorul toolului, dupa cum se vede fleets la 0.025 este el mai mic apoi creste la 0.05, iar creste la 0.082 si apoi scade la 0.1, considerand acest develop as spune ca ne aflam in slow start phase.



Exercitiul 13:

Considerand imaginea anterioara as putea spune ca intr-adevar par sa aiba o perioada, iar aceasta este aproximativ de 0.025 sec.

T1 = 0.025

T2 = 0.053

T3 = 0.082

T4 = 0.1

T12 = T2 – T1 = 0.053 – 0.025 = 0.028

T23 = T3 – T2 = 0.082 - 0.053 = 0.029

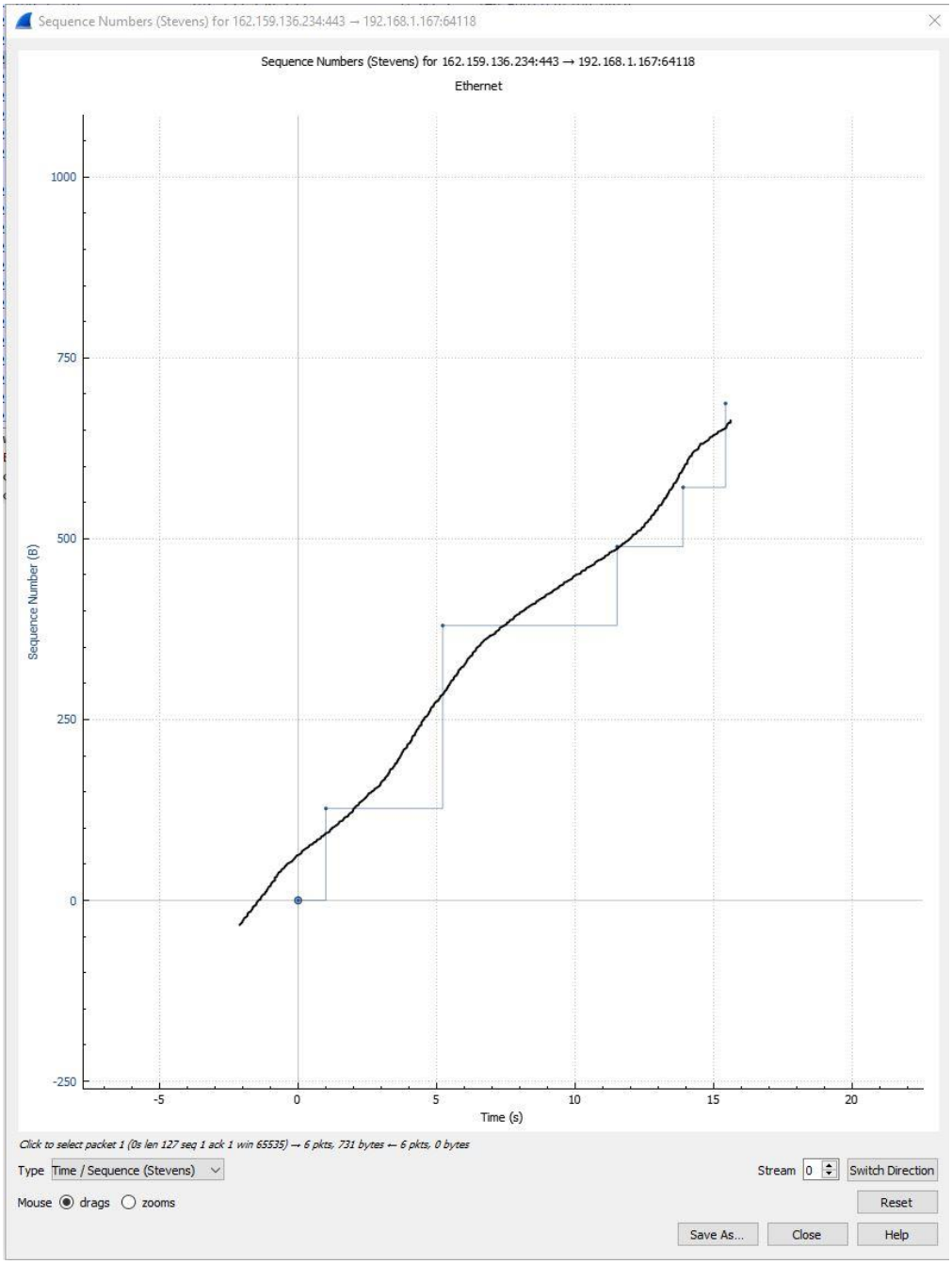
T34 = T4 – T3 = 0.1 - 0.082 = 0.018

Perioada = (0.028 + 0.029 + 0.018) / 3 = 0.025.

Exercitiul 14:

Am capturat pe PC ul meu, dar doar pentru exercitiul 14, restul de exercitii sunt realizate cu captura downloadata.

Pentru prima parte (adica refacerea exercitiului 12): de aceasta data graficul arata ca si cum ar fi in etapa “congestion avoidance”.


Sequence Numbers (Stevens) for 162.159.136.234:443 → 192.168.1.167:64118
Sequence Numbers (Stevens) for 162.159.136.234:443 → 192.168.1.167:64118
Ethernet
Sequence Number (B)
1000
750
500
250
0
-250
-5
0
5
10
15
20
Time (s)
Click to select packet 1 (0s len 127 seq 1 ack 1 win 65535) → 6 pkts, 731 bytes ← 6 pkts, 0 bytes
Type
Time / Sequence (Stevens)
Stream
0
Switch Direction
Mouse
drags
zooms
Reset
Save As...
Close
Help

Iar pentru refacerea exercitiului 13, perioada pare ca acum sa fie aproximativ la 4 5 secunde.

T1 = 1

T2 = 5.2

T3 = 11.5

T4 = 13.8

T12 = T2 – T1 = 5.2 – 1 = 4.2

T23 = T3 – T2 = 11.5 – 5.2 = 6.3

T34 = T4 – T3 = 13.8 – 11.5 = 2.3

Perioada = (4.2 + 6.3 + 2.3) / 3 = 4.26666666667.